

KherveTeX (kherveDOC): A WYSIWYG LaTeX Document Editor with Git-Based Version History

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Abstract

KherveTeX (repository name `kherveDOC`) is an open-source WYSIWYG document editor that produces publication-quality PDFs via LaTeX. Documents are edited on a Word-style paginated surface backed by a typed document model; on every edit the model is serialised to LaTeX and compiled with the `tectonic` engine, giving a live PDF preview. The editor supports inline and display mathematics, figures, tables, citations, cross-references, footnotes, and import of `.docx` and `.tex` files. Every document carries its own Git repository with auto-commit on save and optional push to a remote. KherveTeX is written in Python with PySide6 and released under the GNU General Public License v3.0 at <https://github.com/gkerherve/kherveDOC>.

1 Statement of need

Scientific writing is split between two workflows. Word processors are approachable but offer weak support for mathematics, cross-referencing and consistent typography; LaTeX produces excellent documents but presents a steep learning curve that excludes many students and experimentalists. Bridging tools exist — LyX [1] and TeXmacs [2] pioneered structured WYSIWYG editing, and Overleaf moved LaTeX into the browser — yet there remains room for a lightweight, native editor that (i) looks and behaves like a familiar word processor, (ii) keeps human-readable LaTeX as its archival output, and (iii) records the full history of a document automatically.

KherveTeX addresses this gap. It was developed to let researchers and students who have never written LaTeX produce properly typeset reports, theses and manuscripts, while still emitting clean `.tex` sources that collaborators or journals can consume directly. The built-in Git history removes the “`report_final_v3_REALLY_final.docx`” problem: every save is a commit, and the history viewer allows any previous state to be inspected or restored.

2 Functionality

The editing surface is a Word-style page card rendered at real A4, Letter or Legal dimensions, with dashed page-break lines at every layout pagination. Users write with a paragraph-style picker (Body, Title, Headings 1–5) and inline marks (bold, italic, underline, strikethrough, code, small caps, sub/superscript). Structured inserts include inline and display mathematics (typed as LaTeX), bulleted and numbered lists, hyperlinks, footnotes, citations (`cite/citep/citet`), cross-references (`ref/eqref/pageref`), captioned and labelled figures, tables, page breaks, horizontal rules, and a raw-LaTeX escape hatch for anything not covered by the model.

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Three synchronised tabs show the same document as (1) the formatted WYSIWYG view, (2) live, read-only, syntax-highlighted LaTeX source, and (3) the compiled PDF preview. The native format is `.ktex.json`, a JSON serialisation of the document model, saved alongside a sibling `.tex`; a bundled `.ktexz` ZIP format embeds images. Import is supported from `.tex` (a regex-based subset parser with raw-LaTeX fallback) and from `.docx` via `python-docx`, including extraction of embedded images. Export targets `.tex` and `.pdf`.

3 Implementation

KherveTeX follows a strict model-driven architecture: a typed document tree (Python dataclasses) is the single source of truth. The editor reads and writes the model; a serialiser converts the model to LaTeX; the `tectonic` engine [3] compiles it; and `PyMuPDF` [4] rasterises the resulting pages into the preview pane. No component edits LaTeX strings directly. Version control uses `pygit2` [5]: each document folder is initialised as its own Git repository, saving auto-commits the human-readable `.tex`, and pushes to `origin` on a best-effort basis. The GUI is built with `PySide6` (Qt 6). The test suite covers the document model, the LaTeX serialiser and the importers.

4 Availability

Source code, a Windows installer and documentation are available at <https://github.com/gkerherve/kherveDOC> under the GPL-3.0 license. KherveTeX runs on Python 3.10+ with `PySide6`, `PyMuPDF`, `pygit2` and the `tectonic` binary. It is part of the Kherve family of open-source scientific tools, which includes `KherveFitting` [6], `KherveBook` and `KhervePDF`.

References

- [1] The LyX Team, *LyX — The Document Processor*, <https://www.lyx.org>.
- [2] J. van der Hoeven, *GNU TeXmacs: A free, structured, wysiwyg and technical text editor*, Cahiers GUTenberg 39–40 (2001) 39–50.
- [3] P. K. G. Williams et al., *Tectonic: a modernized, complete, self-contained TeX/LaTeX engine*, <https://tectonic-typesetting.github.io>.
- [4] J. X. McKie, *PyMuPDF: Python bindings for MuPDF*, <https://github.com/pymupdf/PyMuPDF>.
- [5] *pygit2: Python bindings for libgit2*, <https://www.pygit2.org>.
- [6] G. Kerhervé, D. J. Payne, *KherveFitting: An Open Source Software for Fitting X-Ray Photoelectron Spectroscopy Data*, Surf. Interface Anal. (2026), doi:10.1002/sia.70032.